Bihar Engineering University, Patna

End Semester Examination - 2023

Course: B.Tech. | *Semester-IV* | *Time: 03 Hours*

Code: 101405 | *Subject: Introduction to Solid Mechanics* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[2 x 7 = 14]**

(a) In case of a circular section the section modulus is given as :
(i) $\pi d^2/16$ (ii) $\pi d^3/16$ (iii) $\pi d^3/32$ (iv) $\pi d^4/64$

(b) The temperature strain in a bar is _____ proportional to the change in temperature.
(i) Directly (ii) indirectly
(iii) Either (i) or (ii) (iv) None of the above

(c) The ratio of lateral strain to linear strain is known as
(i) modulus of elasticity (ii) modulus of rigidity
(iii) Poisson's ratio (iv) elastic limit

(d) A beam of length 6 m carries a point load 120 kN at its centre. The beam is fixed at both ends. The fixing moment at the ends is
(i) 40 kNm (ii) 90 kNm
(iii)120 kNm (iv) 150 kNm

(e) If the principal stresses on a plane stress problem are σ_1=100 MPa and σ_2= 40 MPa, then the magnitude of shear stress (in MPa) will be
(i) 60 (ii) 50 (iii) 30 (iv) 20

(f) A cantilever beam AB of length *l* has moment M applied at free end. The deflection at the free end B is given as
(i) $M^2 l / EI$ (ii) $M l^2 /2 EI$
(iii) $M l / 2EI$ (iv) $M l^3/2 EI$

(g) Which of the following are usually considered as thin cylinders?
(i) Boilers (ii) Tanks
(iii) Water pipes (iv) All of these

(h) Two shafts in torsion will have equal strength if
(i) only diameter of the shafts is same
(ii) only angle of twist of the shafts is same
(iii) only material of the shafts is same
(iv) only torque transmitting capacity of the shafts is same

(i) Oil tanks, steam boilers, gas pipes are the examples of
(i) Thick shells (ii) Thin cylinders
(iii) Hoop cylinders (iv) Longitudinal cylinders

(j) Two closed coil helical springs of stiffness's K_1 and K_2 are connected in parallel. The combination is equivalent to a single spring of stiffness.
(i) $\sqrt{K_1 K_2}$ (ii) $\frac{K_1+K_2}{2}$ (iii) $K_1 + K_2$ (iv) $\frac{K_1 K_2}{K_1+K_2}$

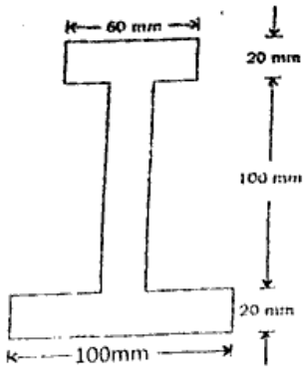
Q.2 (a) Draw stress-strain curve for brittle materials, and show its yield point is determined. **[7]**

(b) A steel rod of 30 mm diameter and 400 mm length ws tested in a testing machine. At a load of 135 kN, the extension in a gage length of 50 mm was measured to be 0.045 mm and the reduction in diameter was 0.008 m. Determine Poisson's ratio and values of three elastic modulii for the test material. **[7]**

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Q.3 (a) A beam of an I-section show in figure is simply supported over a span of 4 metres. Determine the load that the beam can carry per meter length, if the allowable stress in the beam is 30.82 N/mm^2 [7]

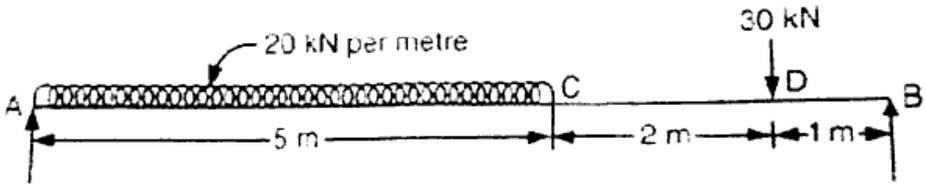


(b) What is Mohr's stress circle? What is the importance of this circle? [7]

Q.4 (a) Define the following terms: [8]
(i) Pure bending (ii) Neutral axis
(iii) Section modulus (iv) Moment of resistance

(b) What do you understand by Strain Energy? Define and derive Castigliano's theorem. [6]

Q.5 A simply supported beam is loaded and supported as shown in the figure given below: [14]



Draw the shear force and bending moment diagrams, and determine the magnitude and location of maximum bending moment.

Q6 (a) Derive the bending equation: $\frac{M}{I} = \frac{\sigma}{y} = \frac{E}{R}$ .What do these symbols mean? State clearly the assumptions. [8]

(b) What do you understand by second moment of area and moment of inertia? [6]

Q.7 (a) With the help of suitable assumptions, deduce torsion equation for a hollow circular shaft. [7]

(b) A hollow circular shaft 20 mm thick transmits 294 kW at 200 r.p.m. Determine the diameters of the shaft if the shear strain due to torsion is not to exceed 8.6×10^{-4}. Assume modulus of rigidity as 80 GN/m^2. [7]

Q.8 (a) Calculate the change in dimensions of a thin cylindrical shell due to an internal pressure. Also calculate the change in length and diameter of the cylindrical shell. [7]

(b) A cylindrical shell 3 m long which is closed at the ends has an internal diameter of 1 m and a wall thickness of 15 mm. Calculate the circumferential and longitudinal stresses induced and also change in dimensions of the shell if it is subjected to an internal pressure of 1.5 MN/ m^2. Take E = 200 GN/m^2 and 1/m = 0.3 [7]

Q.9 A plane element in a body is subjected to a normal stress of 15 kN/m^2 (tensile) in the x – direction as well as a shearing stress of 5 kN/m^2 (clockwise along perpendicular to x-axis). Draw Mohr's circle to determine: [14]

(a) Normal and shearing stress intensities on a plane inclined at an angle of 40^0 to the normal stress.

(b) Principal stresses and their directions

(c) Maximum shearing stress

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